Java Basics – Sunday Class Notes

(First Program, Variables & Input)

Hi everyone,

These notes are to give you an idea of what we will be covering in the class.

You are not expected to learn or memorize everything now. Please read once calmly.

Just read once to get familiar with the terms — in class, I'll explain everything step by step with real-life examples and we will type together during the session.

1. Java Class

In Java, everything is written inside a class.

You can think of a class like a blueprint or design.

Example: Blueprint of a house → Class

Note: Real house → Object (we will learn objects later)

For now, just remember: **A Java program must have at least one class.**

2. main() Method

The main method is the starting point of a Java program.

When we run a Java program, the JVM looks for the main method first and starts execution from there.

You can think of it like: Starting point of a movie / Entry gate of a building.

3. Structure of a Simple Java Program

A basic Java program contains:

• Class

• main method

• Statements inside the main method

We will write and understand this clearly in class using a **Hello World** example.

4. First Java Program (Hello World)

This is the first program every beginner writes.

Purpose:

• To understand program structure

• To see how Java code runs

• To confirm Java setup is working correctly

I will explain: how to write it, how to run it in VS Code, and what happens internally when we click **Run**.

5. Printing Output

To display output on the screen, we use: **System.out.println()**

We will learn:

• Printing text

• Printing values

• Printing messages with variables

6. Difference between print, println, and printf

System.out.print() → prints output in the same line (no new line).

System.out.println() → prints output and moves to the next line.

System.out.printf() → used for formatted output (neat printing).

Format symbols (we'll use slowly): **%d** (integer), **%s** (string), **%f** (decimal).

Note: printf does not automatically go to next line unless we add **\n**.

Simple Summary: print → same line **|** println → next line **|** printf → formatted output

7. Variables

A variable is used to store data.

You can think of a variable like a box / container / labeled storage place.

Example: age, name, salary, isStudent

We use variables because data can change.

8. Basic Data Types (Introduction Only)

We will learn only basic types for now:

• **int** → whole numbers (age, count)

• **double** → decimal numbers (salary)

• **char** → single character

• **String** → text

• **boolean** → true or false

No need to remember rules now — examples will make it clear.

9. Taking Input from User (Introduction)

Programs become more useful when they take input from users.

We will learn:

• Why input is needed

• What is **Scanner**

• How to take simple input like name and age

We will write small programs combining: class + main method + variables + input + output.

Important Note

Do not worry if something feels confusing while reading. Everything will be explained slowly in class.

You can ask questions anytime — there are no silly questions ■

By the end of the class, you will be able to:

- Write a simple Java program
- Understand class and main method
- Use variables
- Take input and print output

We will cover these later (not in this class yet):

- Packages (package statement) & folder structure
- Classpath (what it means and why it matters)
- Command-line arguments (String[] args) in deeper detail